

GCI World

Introduction to Data science



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1

Why Study Data Science?

2

Data Science as Science

3

What GCI Offers

1

Why Study Data Science?

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Data Science as Science

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What GCI Offers

“Why should I study data science?”

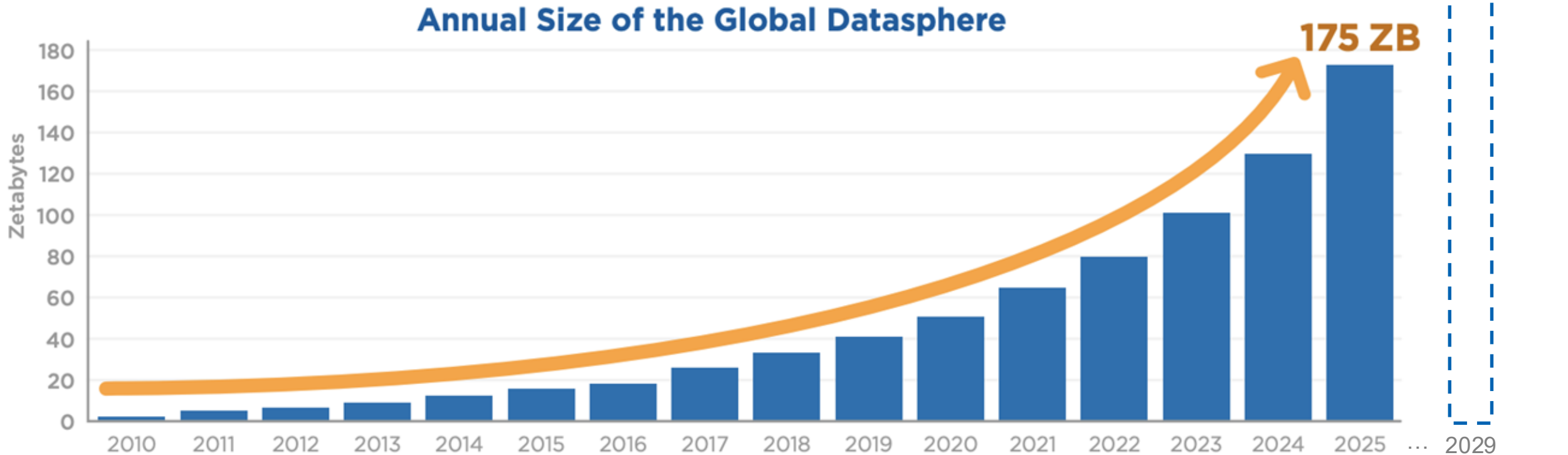
“Because data is everywhere, no matter what field you study. Learning data science helps you find patterns, test ideas, and make better decisions using evidence.”

by ChatGPT

Data is Growing Exponentially



1 ZB (Zettabyte) =
1,000,000,000,000,000,000,000 bytes

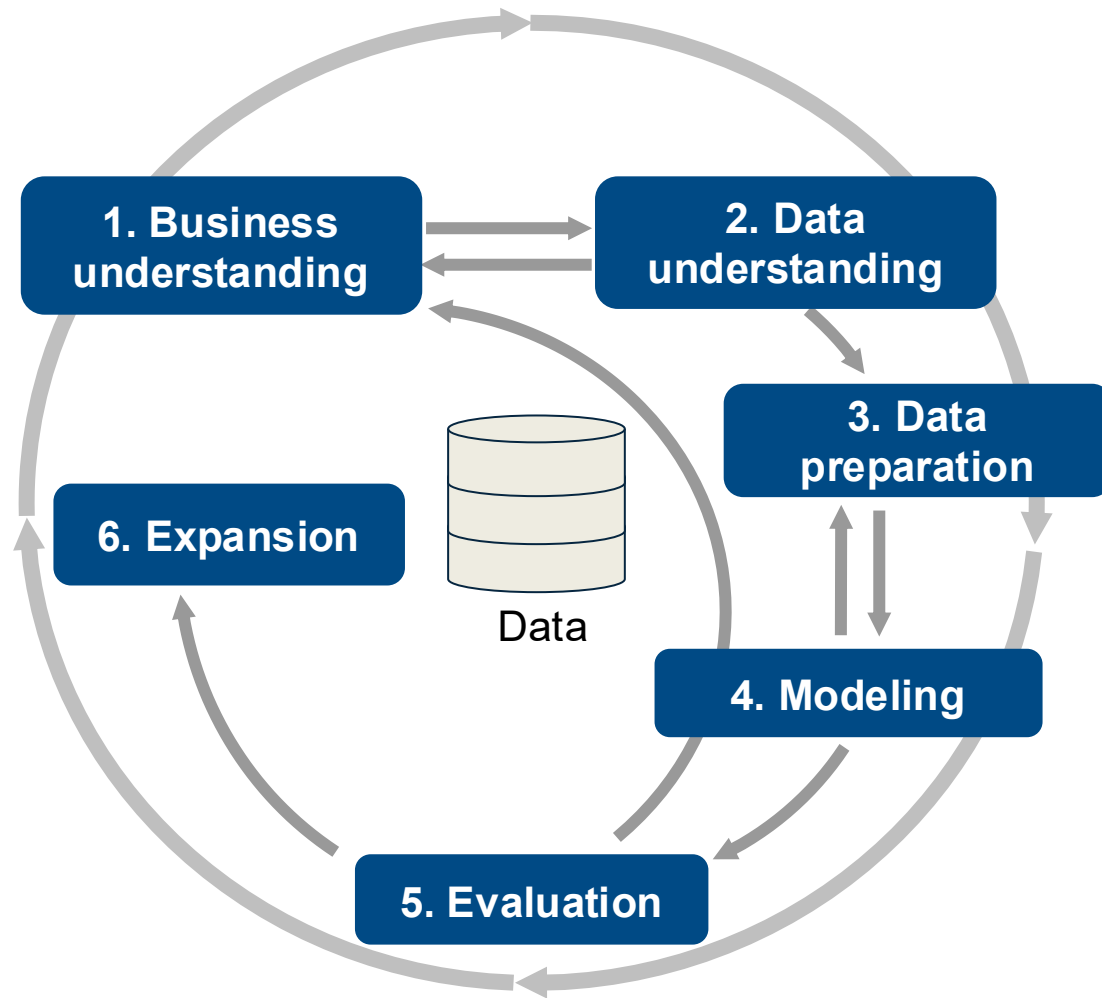


Source: Data Age 2025, sponsored by Seagate with data from IDC Global DataSphere, Nov 2018

Data Already Shapes Decision Across Industries



More Data Does Not Guarantee Better Knowledge



e.g. Food trucks



Yesterday, the food truck sold 40 sandwiches.
We sold all the sandwiches.



When did we sell out?
Is this pattern repetitive?
How are the buying patterns of the customers?



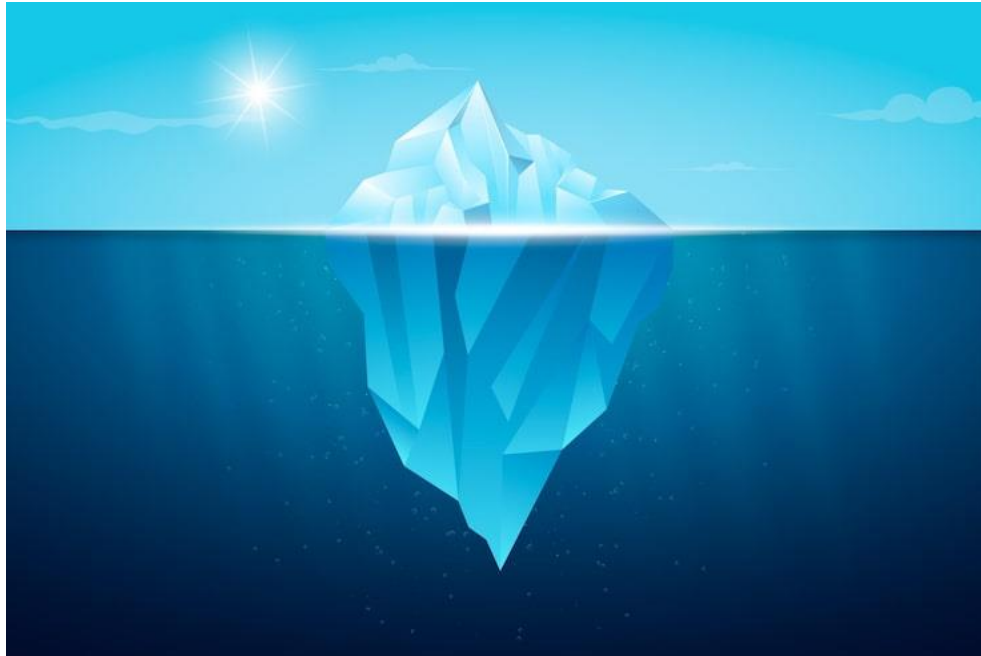
Order more? Discount late? Change opening hours?

Data matters when it leads to actions

Ask What is Missing



Relying solely on the data you have can lead to incorrect decisions.



Visible data (database, file, etc)

Hidden data (abstract concepts including those that are not available to make data out of)

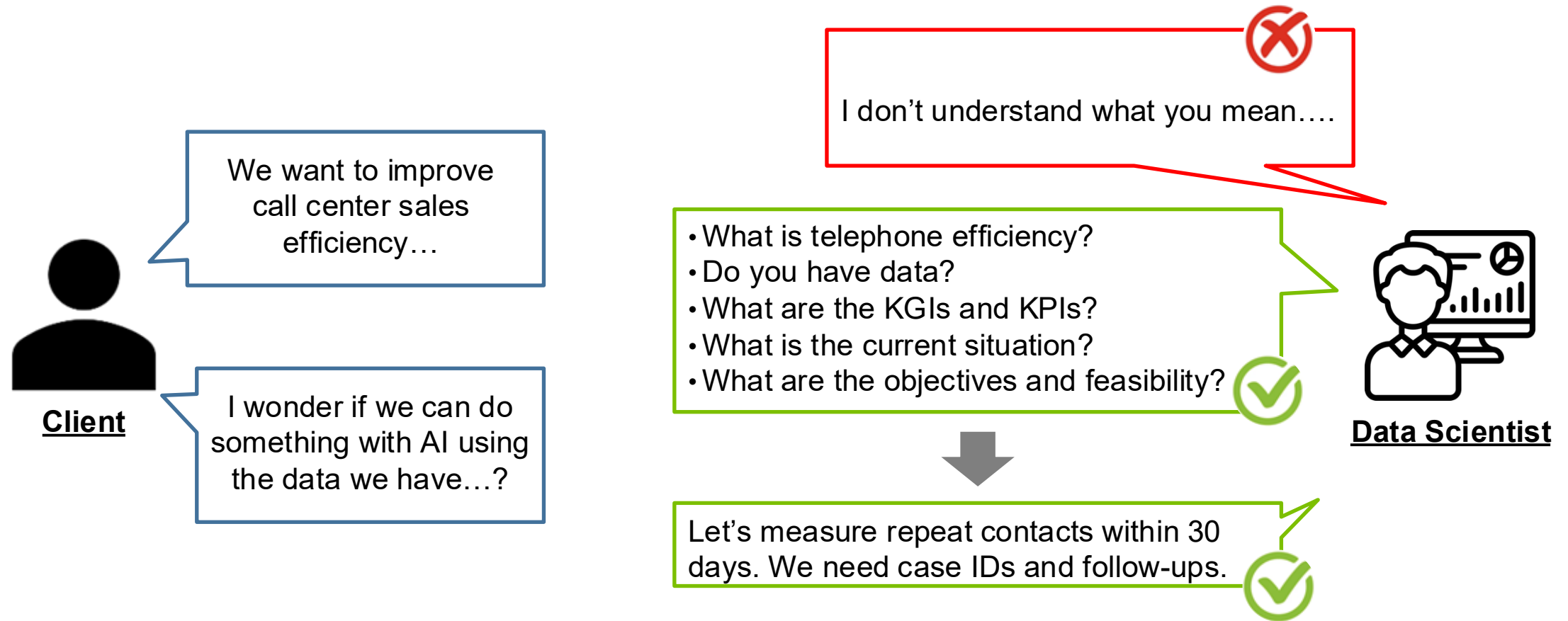
Reference:

“Dark Data”, David J. Hand

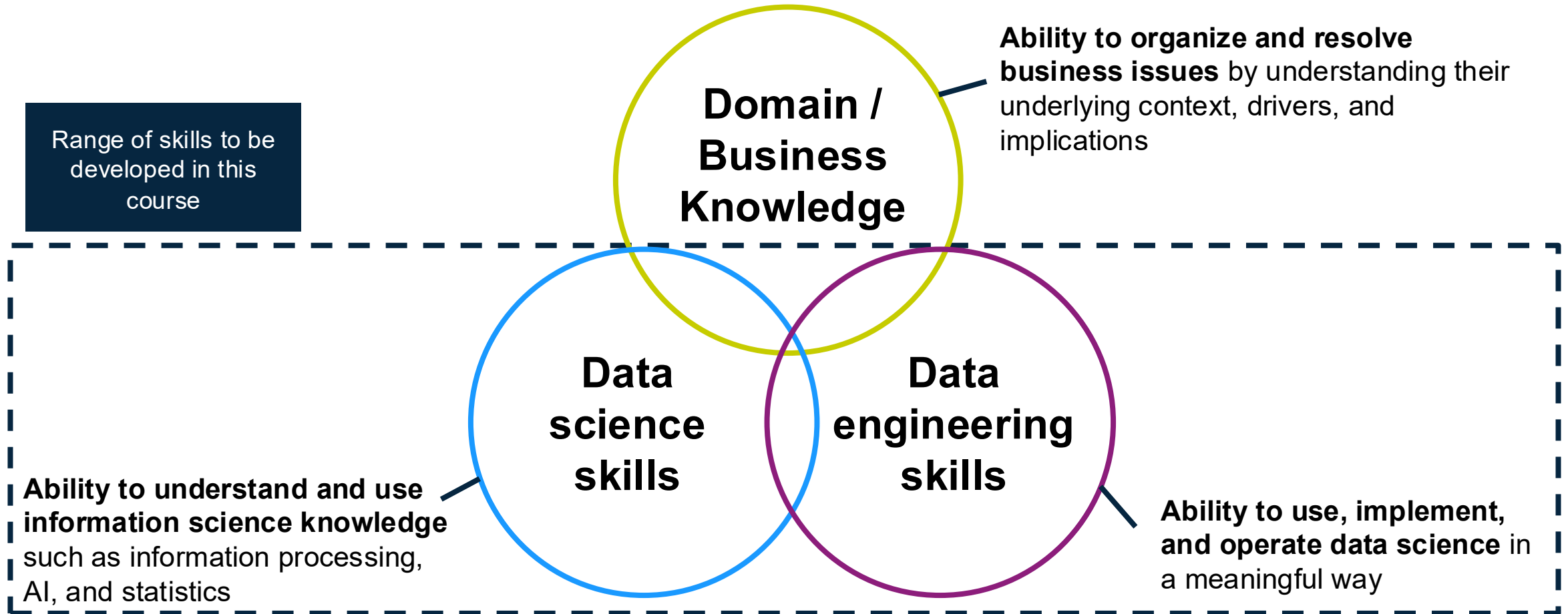
“Mienai mono wo miru (See the unseen)”, Isao Hosoya

Someone Must Connect the Problem to the Data

Define the problem first. Then, translate the problem into something that can be checked.



Skill Set of a Data Scientist



1

Why Study Data Science?

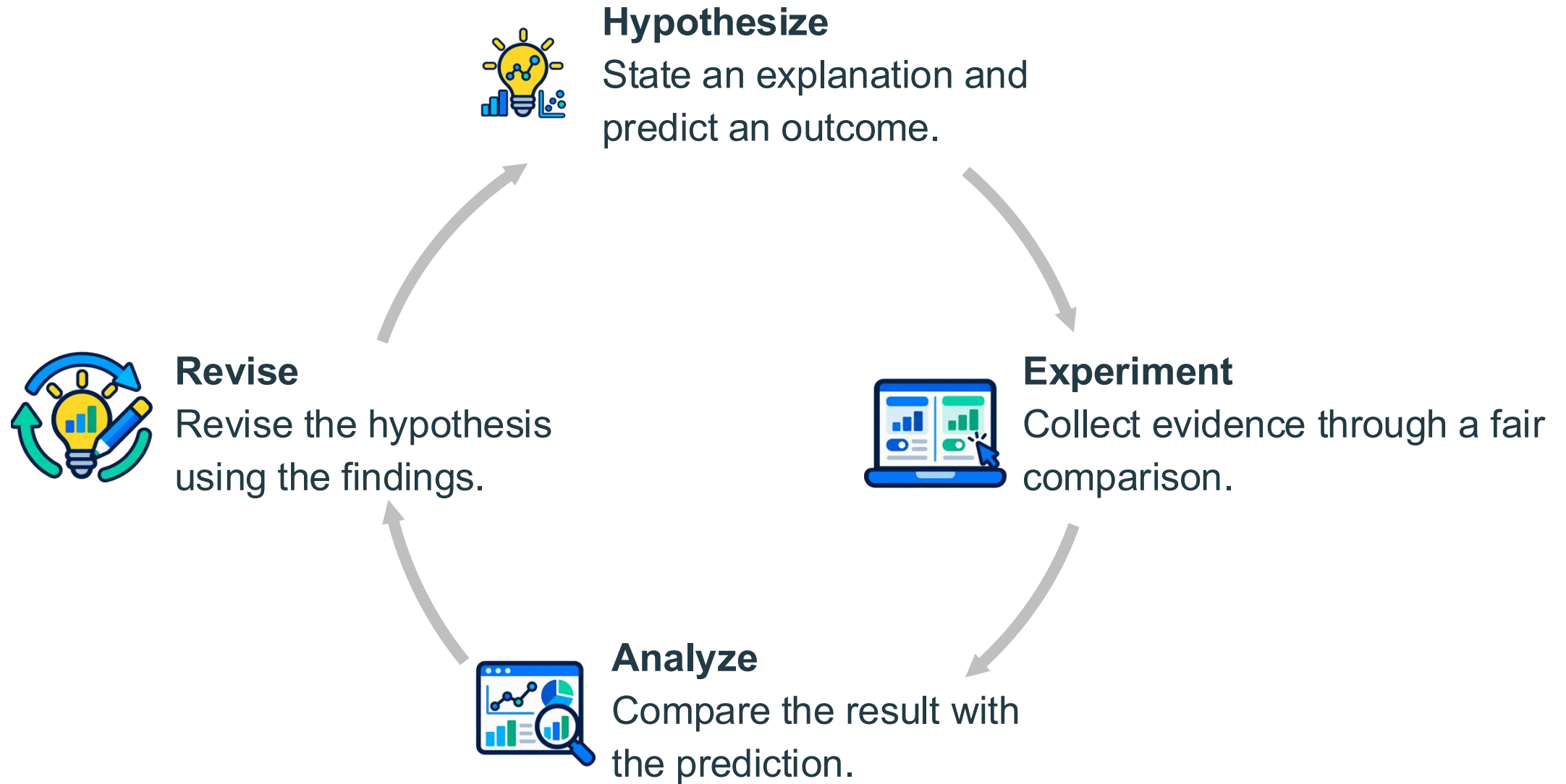
2

Data Science as Science

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What GCI Offers

Data Science Runs the Same Loop as Science



Predict demand, place an order, check sales and use the result to improve the next order.

Seven-Eleven Japan · item-by-item management

Observe

Combine recent sales,
weather and local events.

Hypothesize

Predict demand for one item,
at one store and time.

Repeat

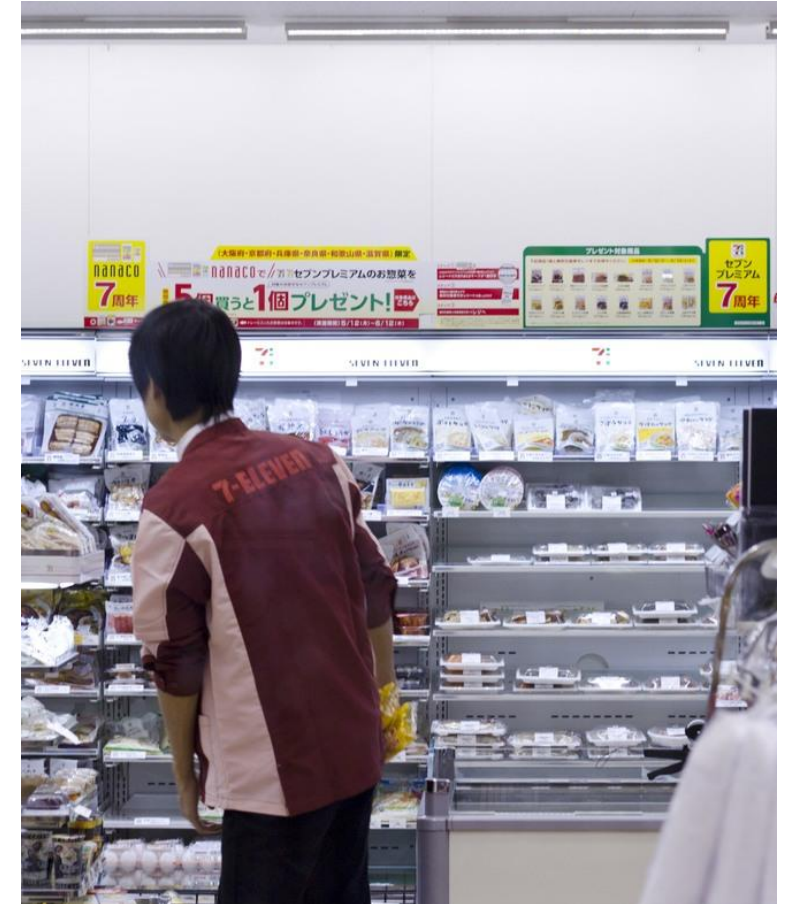


Check & revise

Compare forecast and sales;
revise the next order.

Order & sell

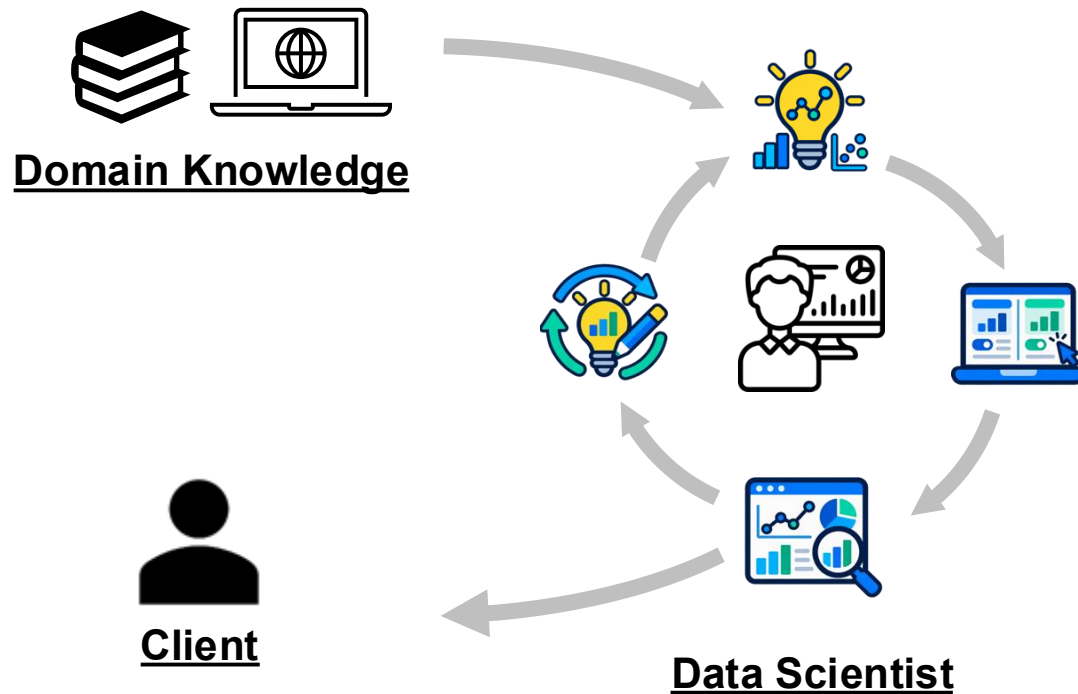
Choose the order quantity
and arrange the display.



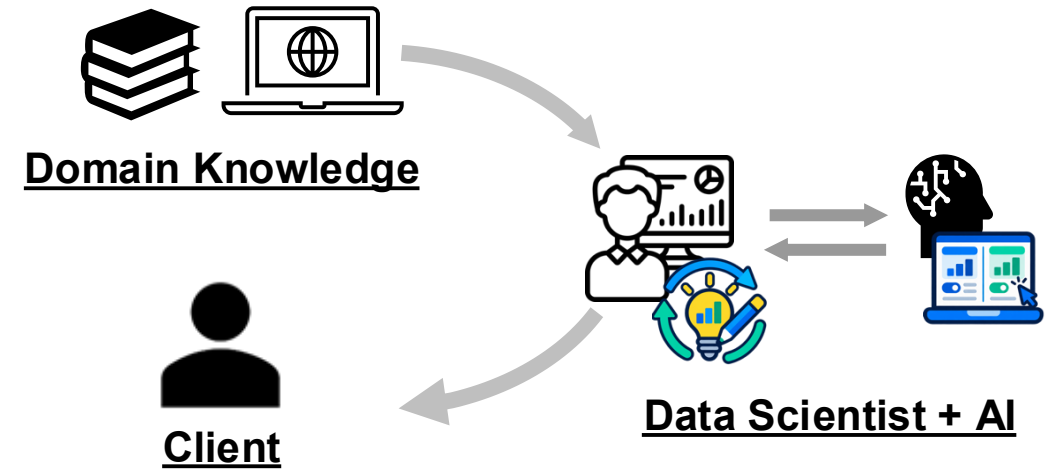
AI Coding Accelerates Iteration



Data science cycle in the past



Data science cycle today



Translating your problem, building hypotheses, and actually solving the problems is still your task.



“Asking the right question is the hardest part of science.”
by Demis Hassabis

3 Keys in Data Science



01

Hypothesis-Driven Objectives



Problem solving in business

02

Actionable Insights



New projects, ideas

03

Methods Comparison



Statistics, data science

“Why Should I Study Data Science?”



Data Science

=

Data

+

Science

Solve real-world **problems** around you and extracting **value**
by **scientifically** analyzing data-driven **insights**.

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What GCI Offers

**To acquire the basic technical skills
necessary for utilizing data in business**

01

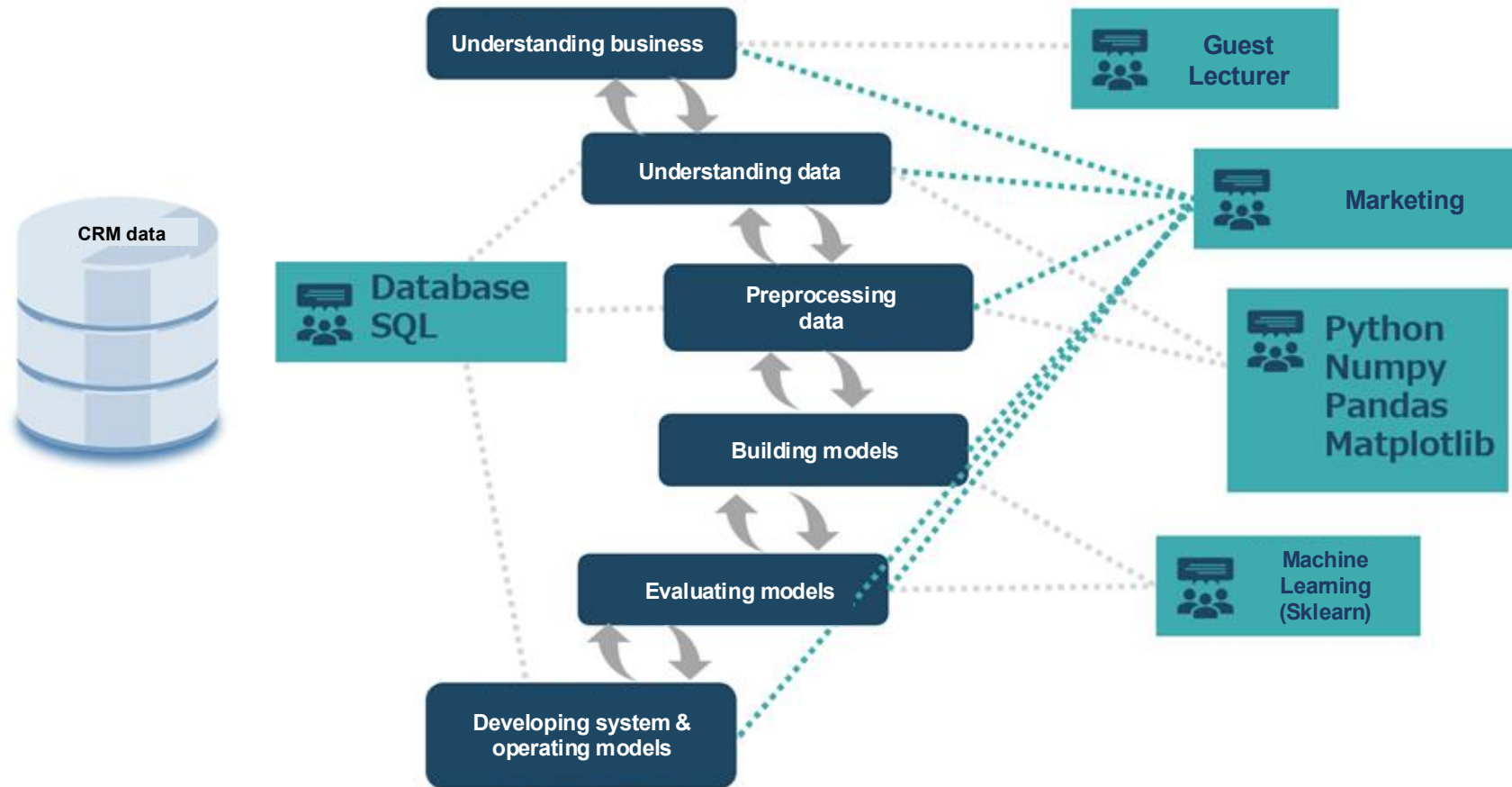
Basic programming

02

Fundamental skills required for data science

03

Marketing mindset for business



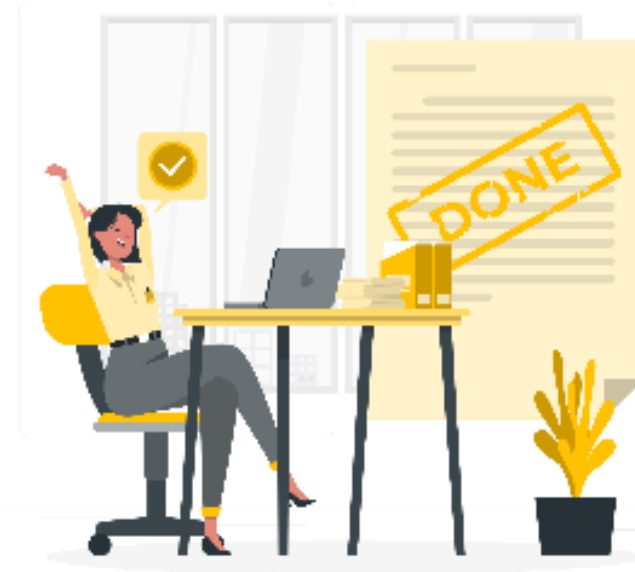
01. Theory

02. Implementation

03. Exercises

04. Explanation of the exercises

(05. Homework)



Don't Try to Remember Everything



Utilize shortcut

It's like swinging practice



Learn at your own pace



On Slack! Use the global community wisely :)

Ask & teach each other

Teaching can also help us understand what we lack understanding on



Research skill

Using google, ChatGPT etc



Example Usage of Generative AI To Enhance Your Study



e.g. When you're stuck on an exercise...

If you don't know what to do



**Ask for the explanation of
the words/concepts**

If you're stuck on an error



**Show AI the error message
(maybe the input data too)**

Example Usage of Generative AI To Enhance Your Study



Quri: AI adaptive learning platform developed by Matsuo Lab

Quri

+ New chat

Enrolled courses

- > GCI World 2026 September
- > KOSEN AI Bootcamp 2026

用語を解説して

Course

GCI World 2026 September

Ask about this course...

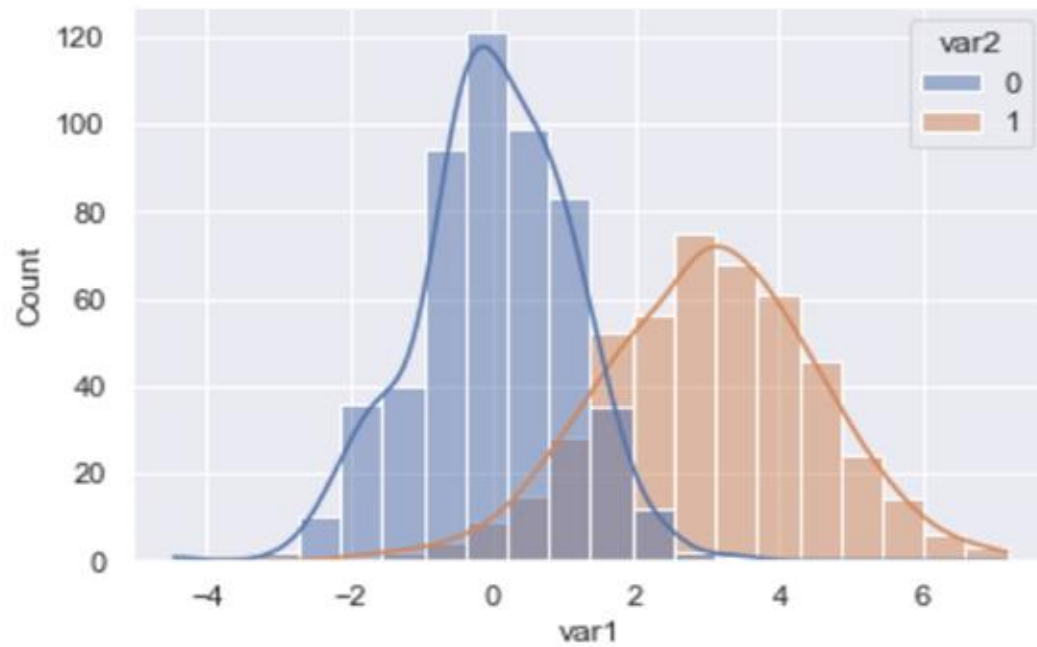
Lecture schedule

9/17 (Thu) 20:00	01 Introduction to Data Science	Up next
9/24 (Thu) 20:00	02 Manipulating Data Using Numpy	
10/1 (Thu) 20:00	03 Cleaning Data Using Pandas	
10/8 (Thu) 20:00	04 Visualizing Data Using Matplotlib	
10/15 (Thu) 20:00	05 Supervised Learning	
10/22 (Thu) 20:00	06 Model Evaluation	
10/29 (Thu) 20:00	07 Competition & Final Assignment Tutorial	
11/5 (Thu) 20:00	08 Feature Engineering	
11/12 (Thu) 20:00	09 Marketing and Data Science	
11/19 (Thu) 20:00	10 SQL	
11/26 (Thu) 20:00	11 Unsupervised Learning	
12/3 (Thu) 20:00	12 Time Series Analysis	
12/10 (Thu) 20:00	13 Guest Session	

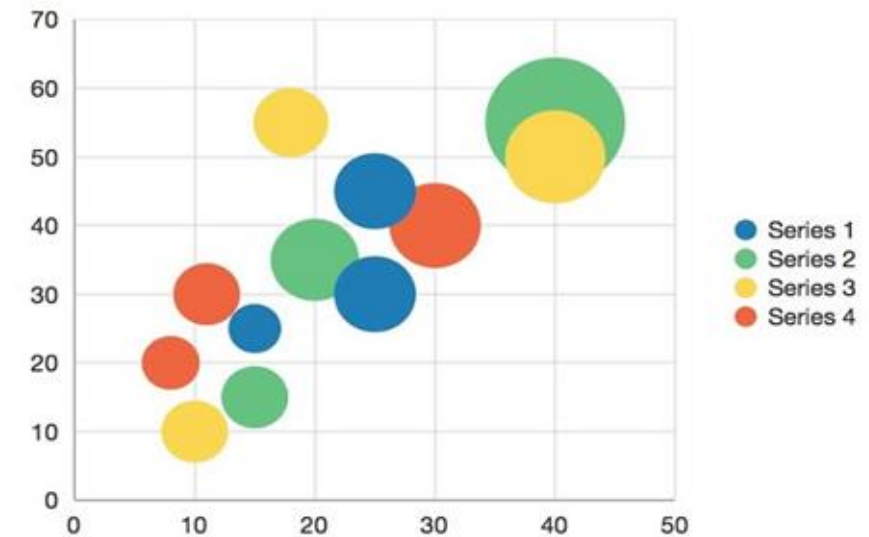
quri.omnicamp.us

- No prior knowledge is strictly required
- Ideal if you know:
 - Elementary calculus and linear algebra
 - Basics of probability and statistics
 - Basic usage of Python, Jupyter notebook, Linux
- Have a PC
 - To do the assignments, a PC is required (no smartphones)
 - We will use Google Colaboratory in GCI so you do not need to install Python yourself
- Make sure to **do some self-study**

Data analysis and visualization methods you should use daily!



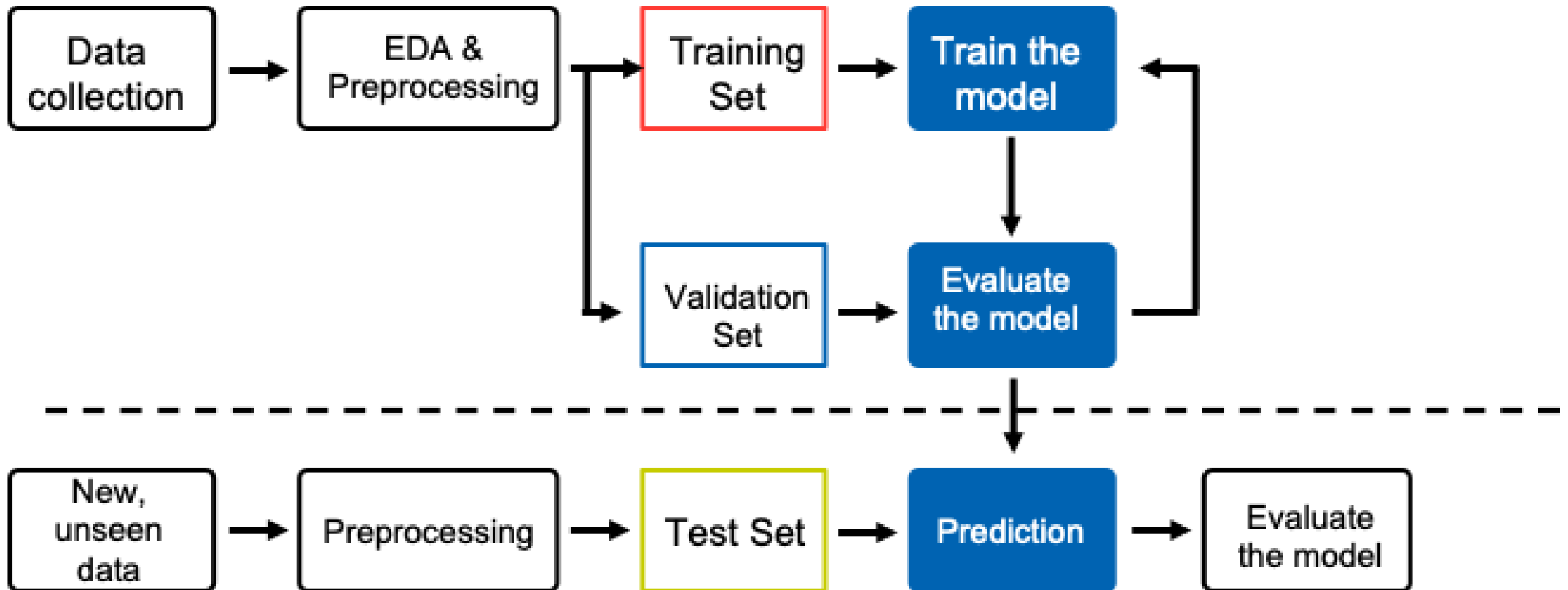
Bubble Chart



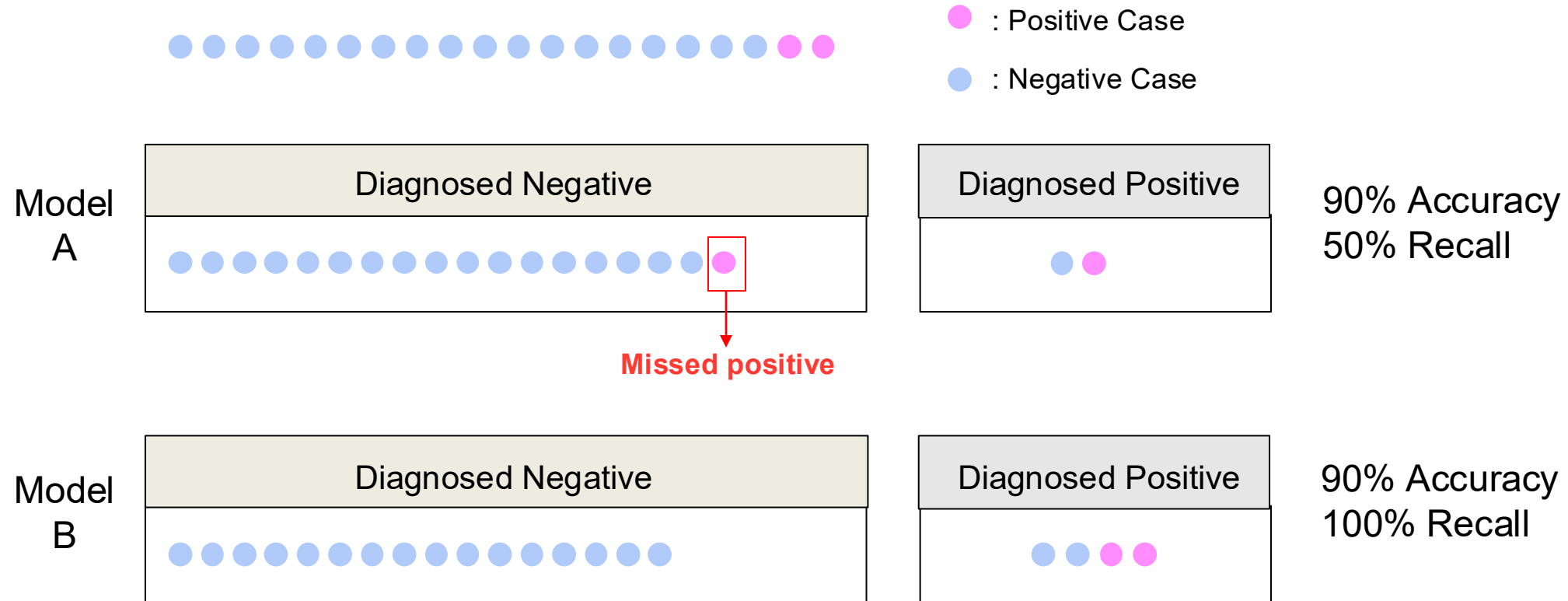
Topics in Week 5



Use how to apply supervised learning to build a model to solve detection and prediction problems.



Model evaluation and feature engineering to get you prepared to become a competitive AI engineer!

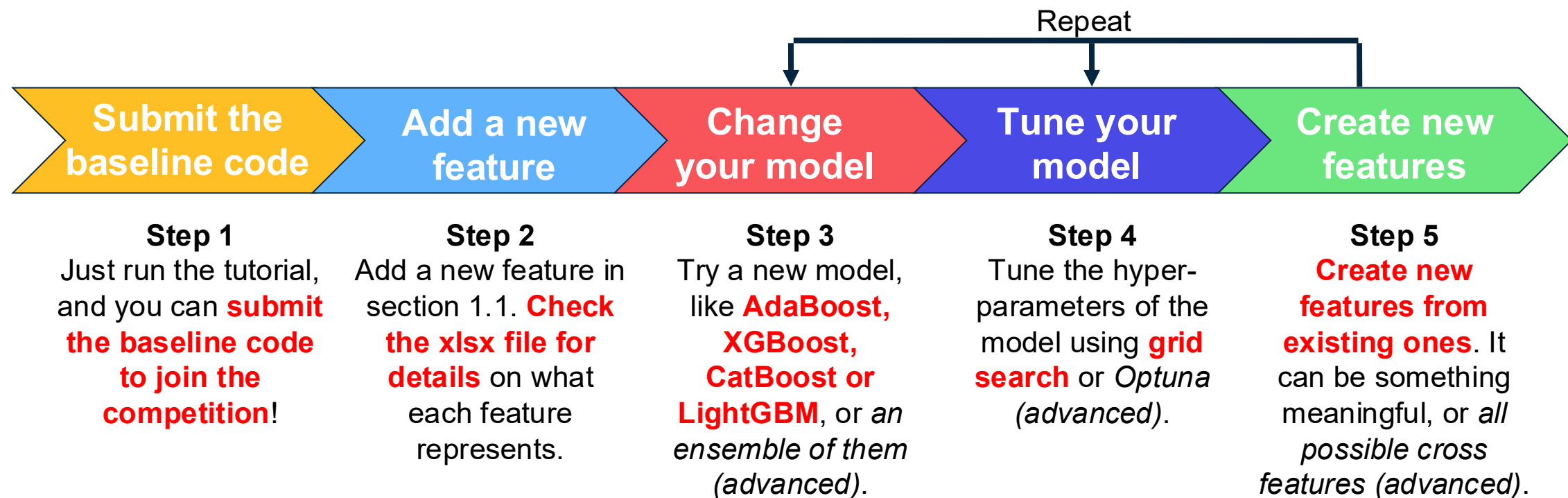


If missed positives cost more, Model B is preferable.

Topics in Week 7



Get started in solving real-world business problems through competition and final assignments.



Topics in Week 9 and 13



Learn how data engineering skills can be translated to real-world application through marketing.
Learn from guest entrepreneurs about actual business cases.

1. Clear, Measurable Goals (KPIs & KGIs):

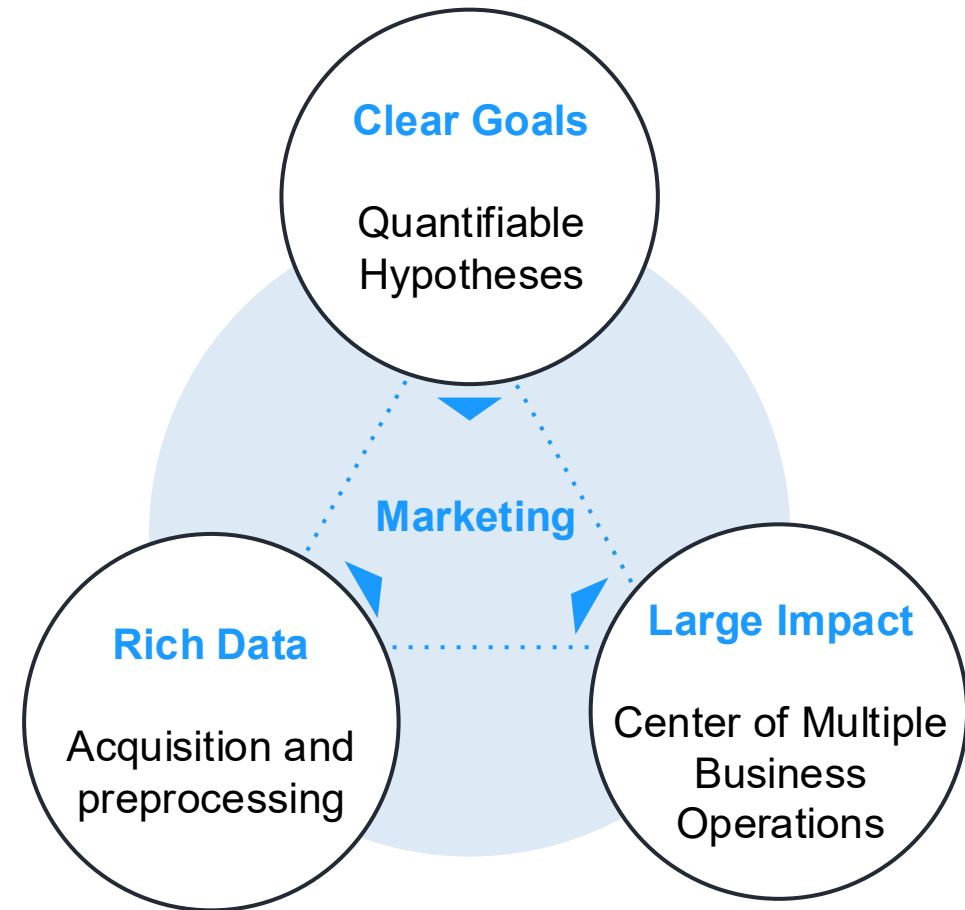
- **Well-defined, measurable goals** (increasing sales, customer retention, reducing churn, and improving customer engagement etc.)
- These goals naturally translate into **quantifiable hypotheses**

2. Rich Availability of Data:

- Inherently produces **vast quantities of data**, such as:
 - Behavioral, Transactional, Demographic data, etc.
- An ideal domain for practicing data acquisition and preprocessing skills.

3. Center of many business operations:

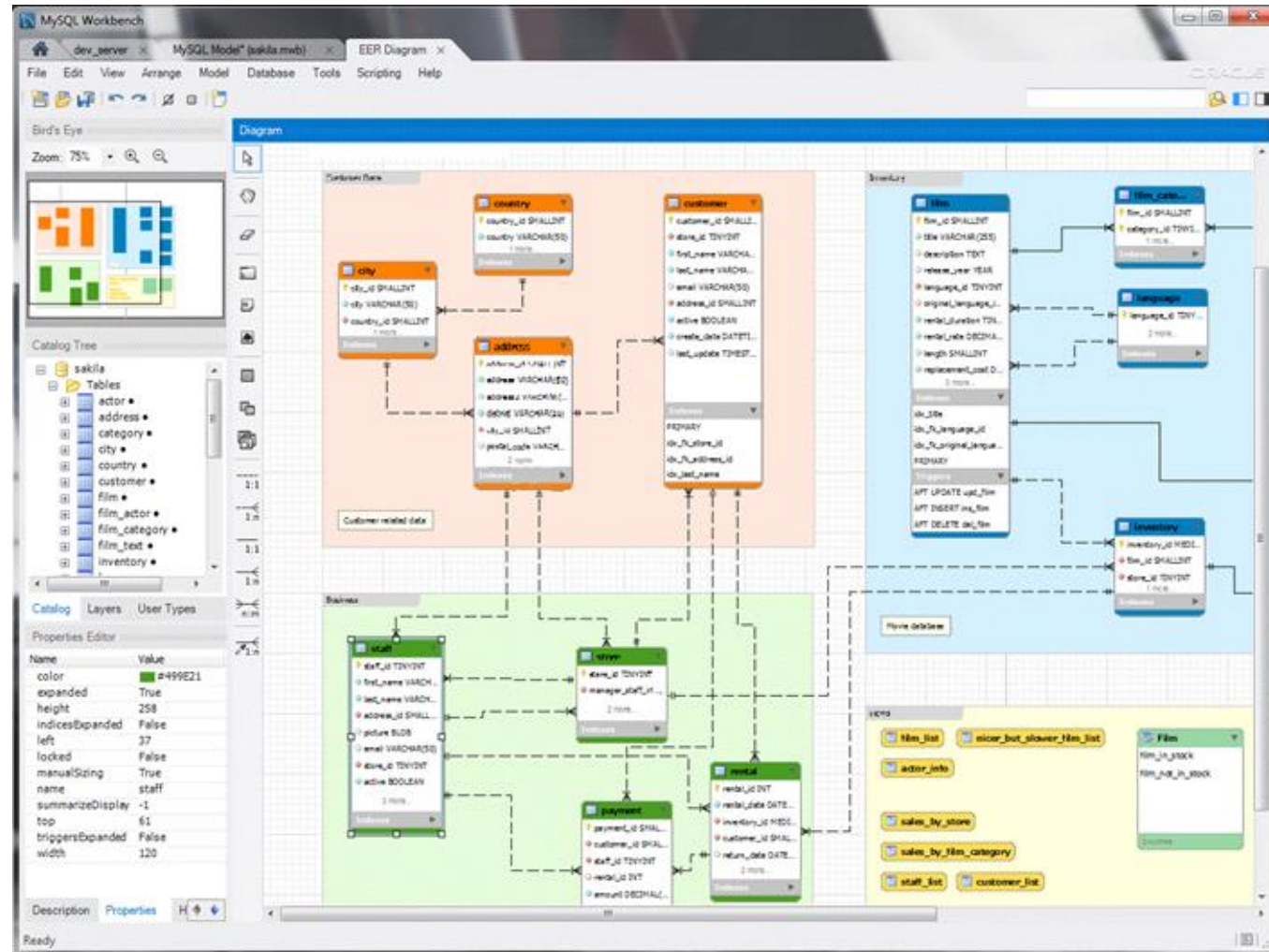
- Often serves as a hub connecting various departments such as R&D, Sales, Customer Support, and Supply Chain Management
- **Hence holding a large impact in business scenes**



Topics in Week 10



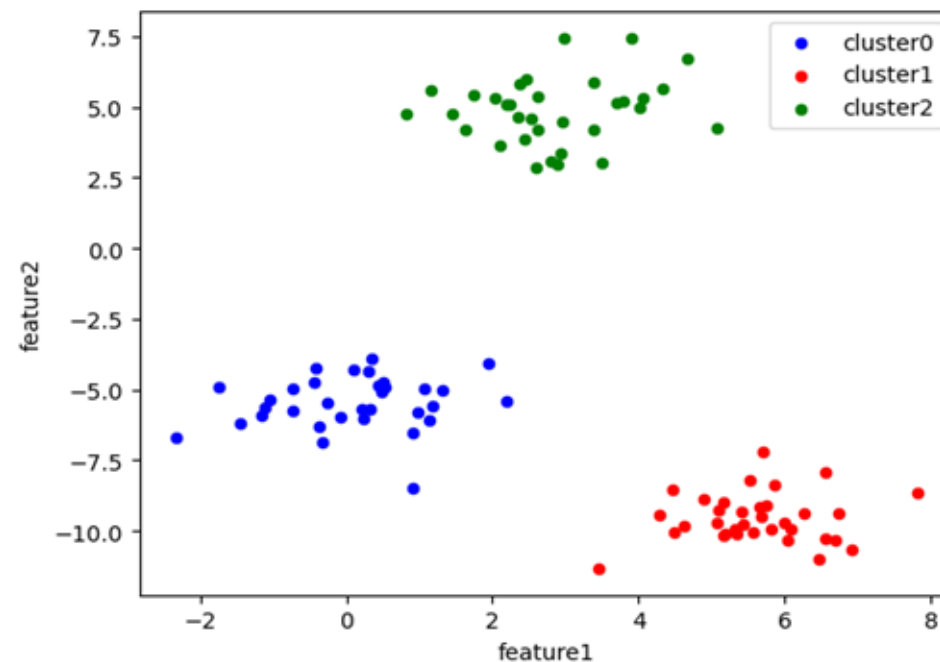
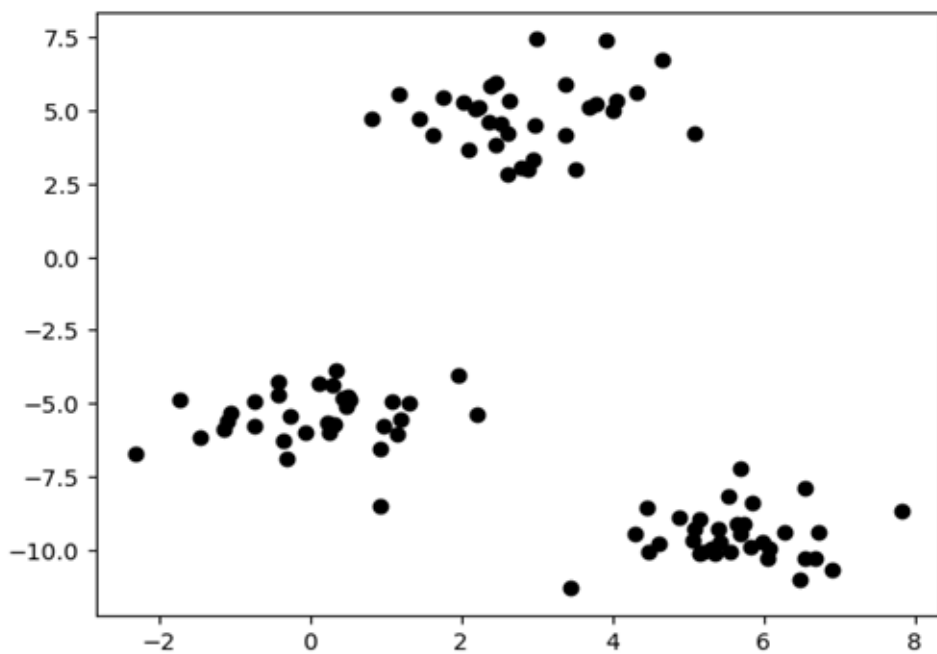
Learn how to use large databases with SQL



Topics in Week 11



Learn how to uncover hidden patterns and structures in unlabeled data



Topics in Week 12



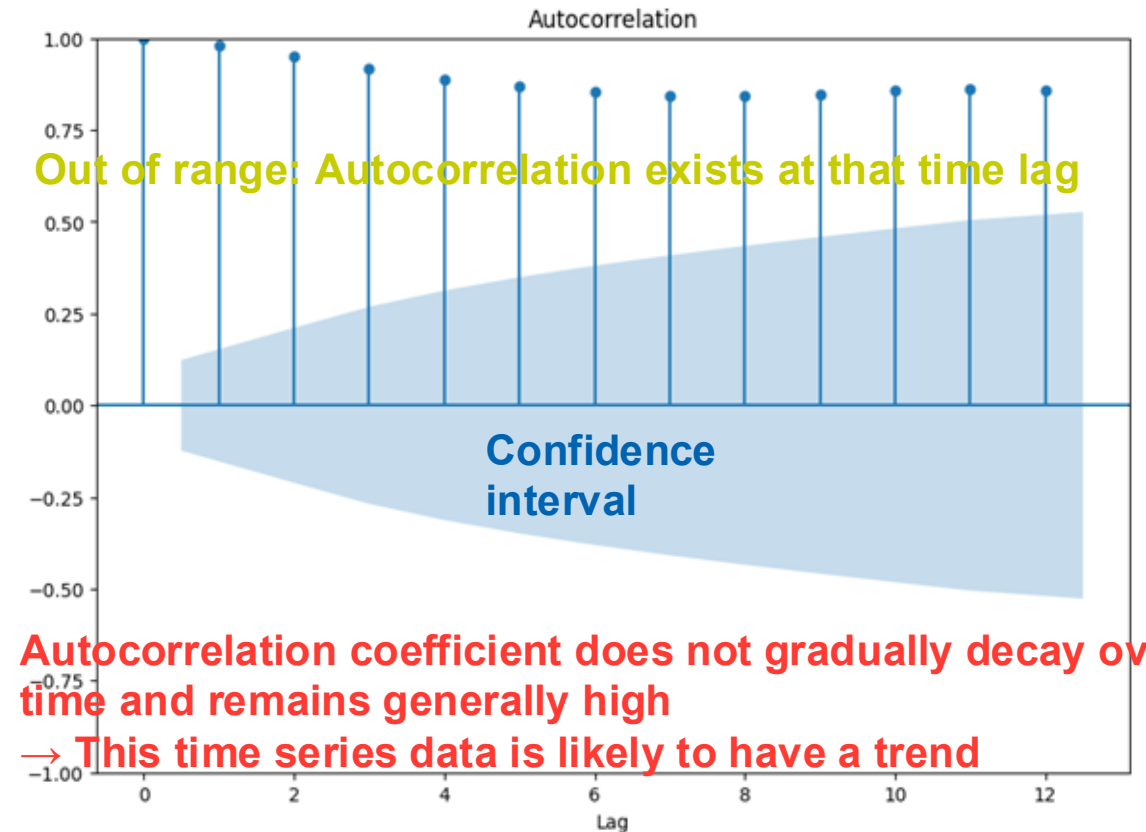
Learn how to analyze time-dependent data, identify patterns and trends, and use them to understand and forecast future behavior.

$$\rho_k = \frac{\sum_{t=k+1}^n (X_t - \bar{X})(X_{t-k} - \bar{X})}{\sum_{t=1}^n (X_t - \bar{X})^2}$$

Calculating the autocorrelation coefficient reveals how much past data influences current data.

- Range: -1 to 1
- Value is closer to 1 = Positive correlation
- Value is closer to -1 = Negative correlation
- Value is around 0 = No correlation

Correlogram plots the autocorrelation coefficients with lag as the x-axis.

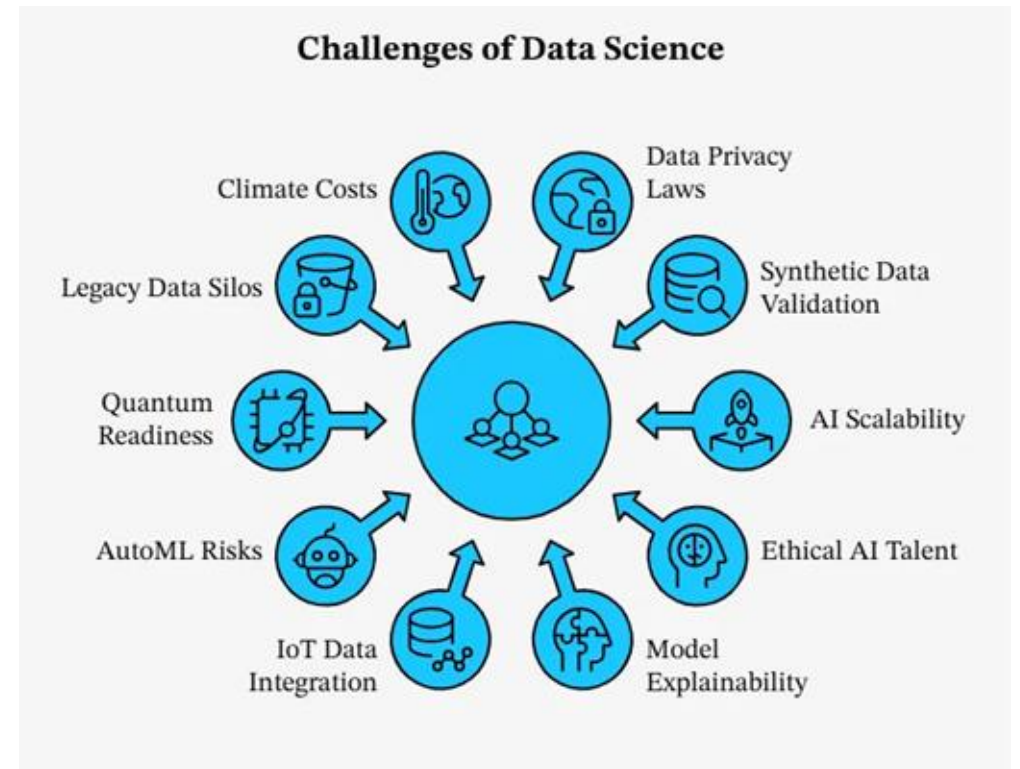


Your GCI Journey Starts Today



- Think about how you might use data science to solve your problem of interest. **You are the expert.**
- **Start with a hypothesis.** Test it. Analyze it.
- **Translate findings into actions.**
- Make sure to **compare results.**
- **Use AI wisely.** At the end of the day, it is your job to understand the analysis and communicate with others.

And always...
discuss with your peers!



https://ik.imagekit.io/upgrad1/abroadimages/imageCompo/images/10_Common_Challenges_of_Data_Science_visual_selectionTKXI9Z.png?pr=true

